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MID-TERM PROJECT ON

FEDERAL EXPENDITURES ON CHILDREN IN 2018 BY U.S.A.

1. INTRODUCTION:

"In a period where data is a key factor in communication and decision-making, the capacity to effectively communicate through data visualization is essential. This report details an investigation into the practice of data visualization and its function in clearly communicating information.
The project's focus is the analysis and redesign of a bad graph with the intention of enhancing its communication capabilities. In this analysis, we will examine the introductory graph and talk about its source, context, advantages, and disadvantages. We will then present the revised graph and go into detail about the goals, additional data sources, and software solutions used. Finally, while maintaining accepted design principles, we will discuss the difficulties encountered and the advancements made.
This report aims to demonstrate our proficiency in data visualization and our capacity for clear graphical communication. By the time this report is finished, we hope to have not only demonstrated an effective redesign but also emphasized how crucial thoughtful data visualization is when presenting complex information.

2. Bad Graph:

- **Source and Context:**
  The bad graph and the related data are taken from the website [Budgeting for the Children: Where Does the Money Go? (howmuch.net)]()
  Author: Irena
  Published: 7th Oct 2019.

  The context of the data relates to the allocation of federal government expenditures for various categories and subcategories associated with children in the year 2018. The data is organized in a tabular format with the following columns:

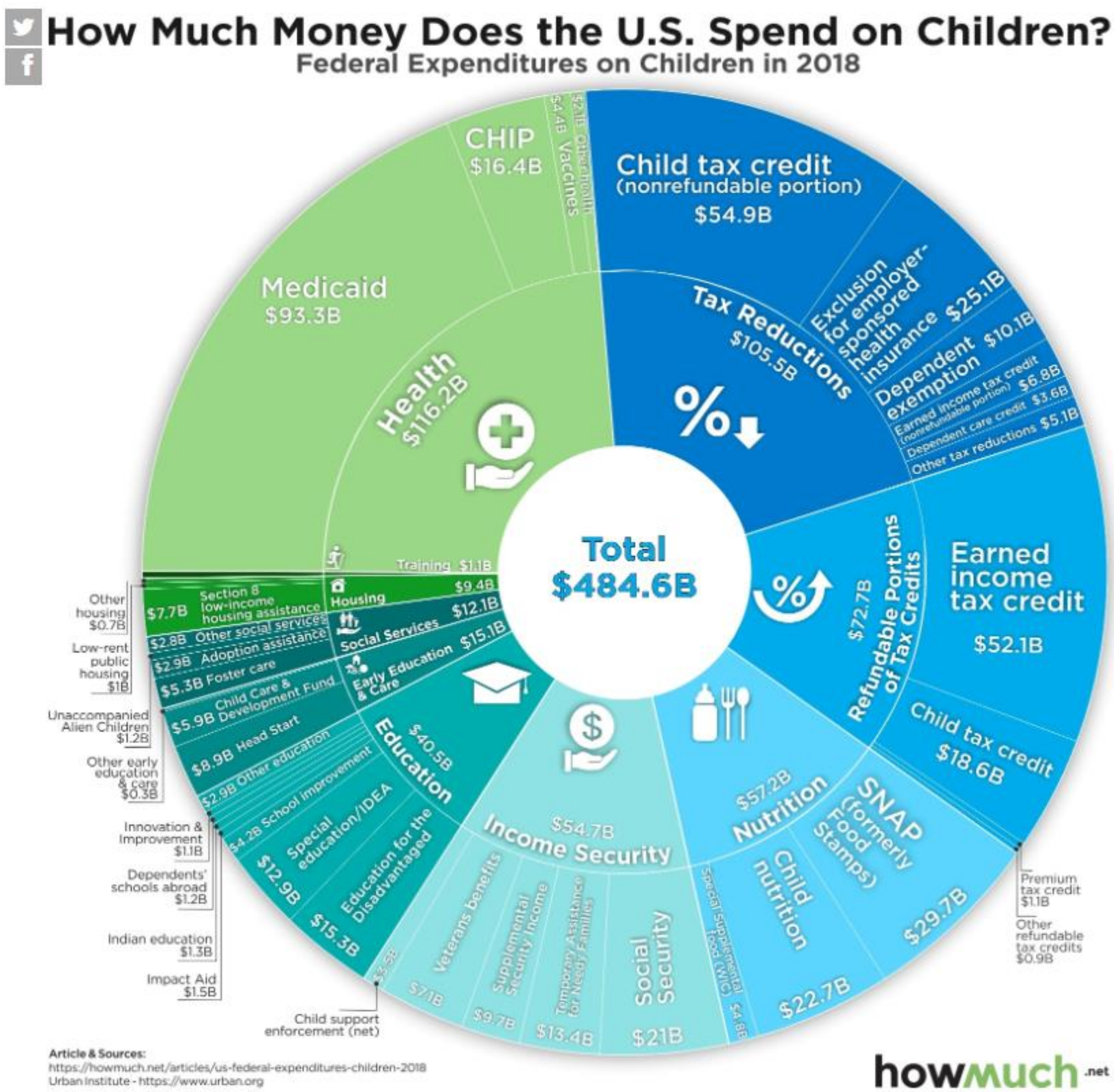
  1. Categories: This column likely contains areas where the federal government allocated funds to support children's well-being. Examples of such categories may include health, nutrition, income security, education, and other related areas.

  2. Subcategories: This column further breaks down the expenditure data into more specific subcategories within each of the categories. For instance, under the "Health" category, you may find subcategories such as Medicaid, Vaccines for Children, and other related areas.

  3. 2018 (Billions of 2018 Dollars): This column likely contains the actual federal expenditures for each category and subcategory in the year 2018. The values are in billions of 2018 dollars, indicating the financial commitment made by the federal government to each program or area for that particular year.

4. Change from 2017: This column likely shows the change in federal expenditures for each category and subcategory from the previous year (2017). It may be expressed as an increase or decrease in funding compared to the previous year.

BAD GRAPH:



DATA:

	2018 (Billions of 2018 dollars)	Change from 2017
1. Health	116.2	2.0
Medicaid	93.3	1.6
CHIP	16.4	0.6
Vaccines for children	4.4	-0.1
Other health	2.1	-0.1
2. Nutrition	57.2	-1.8
SNAP (formerly Food Stamps)	29.7	-1.4
Child nutrition	22.7	-0.1
Special Supplemental food (WIC)	4.8	-0.3
3. Income Security	54.7	-1.1
Social Security	21.0	-0.2
Temporary Assistance for Needy Families	13.4	0.3
Supplemental Security Income	9.7	-1.1
Veterans benefits	7.1	-0.2
Child support enforcement (net)	3.5	*
Other income security	*	*
4. Education	40.5	-1.9
Education for the Disadvantaged (Title I, Part A)	15.3	-1.3
Special education/IDEA	12.9	*
School improvement	4.2	-0.3
Impact Aid	1.5	-0.1
Indian education	1.3	0.1
Dependents' schools abroad	1.2	*
Innovation and improvement	1.1	-0.1
Other education	2.9	-0.3
5. Early Education and Care	15.1	-0.1
Head Start (including Early Head Start)	8.9	-0.2
Child Care and Development Fund	5.9	0.1
Other early education and care	0.3	*
6. Social Services	12.1	0.5
Foster care	5.3	0.3
Adoption assistance	2.9	0.4
Unaccompanied Alien Children	1.2	-0.2
Other social services	2.8	*

6. Social Services	12.1	0.5
Foster care	5.3	0.3
Adoption assistance	2.9	0.4
Unaccompanied Alien Children	1.2	-0.2
Other social services	2.8	*
7. Housing	9.4	0.2
Section 8 low-income housing assistance	7.7	0.1
Low-rent public housing	1.0	*
Other housing	0.7	*
8. Training	1.1	-0.2
9. Refundable Portions of Tax Credits	72.7	-2.9
Earned income tax credit	52.1	-2.2
Child tax credit	18.6	-1.2
Premium tax credit	1.1	0.5
Other refundable tax credits	0.9	*
10. Tax Reductions	105.5	-6.9
Child tax credit (nonrefundable portion)	54.9	24.9
Exclusion for employer-sponsored health insurance	25.1	-0.3
Dependent exemption	10.1	-31.1
Earned income tax credit (nonrefundable portion)	6.8	-0.1
Dependent care credit	3.6	-0.1
Other tax reductions	5.1	-0.3
TOTAL EXPENDITURES ON CHILDREN	**484.6**	**-12.2**
OUTLAYS SUBTOTAL (1–9)	379.0	-5.3

- **STRENGTHS AND WEAKNESSES:**

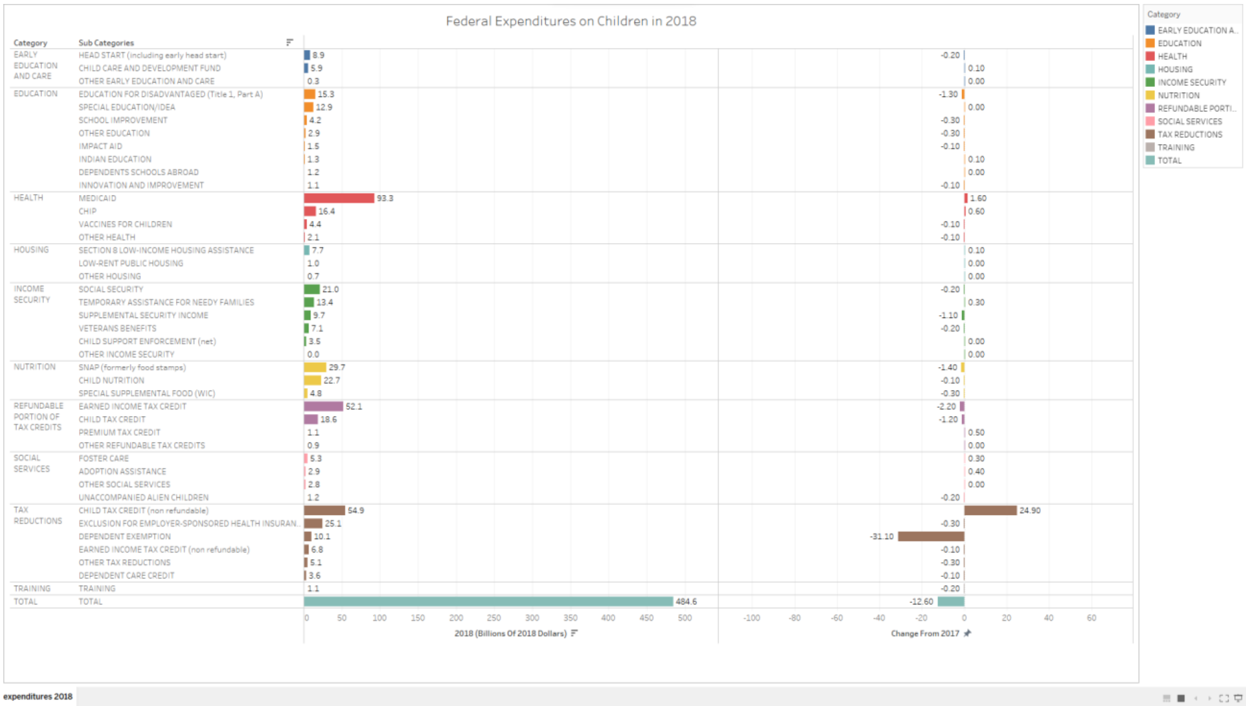
STRENGTHS: We can tell from the above graph that different colors were used to distinguish between the various sectors. Each segment of the data is covered, and the total is displayed in the center with a large font size to help the audience understand. Provides a clear summary of the data, Suitable for visualizing large data sets.
WEAKNESSES: The audience cannot fully comprehend every category and subcategory when they see it. If we clearly observe the graph, we can see that there is a small amount of bias for the areas ($ in Billions) with fewer values, the person should concentrate on the graph to get every single value for each category. The graph is complex and can be shown in a simple way. We know that when we simplify something, we comprehend it more quickly.

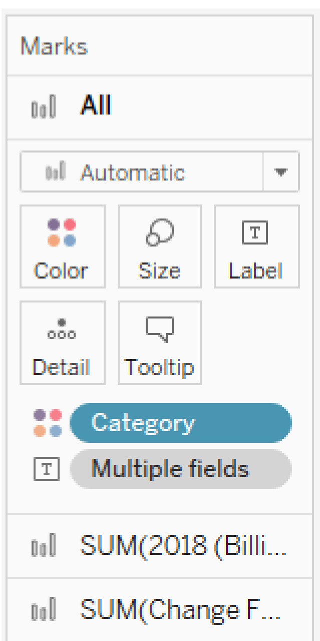
3. **REDESIGNED GRAPH:**

- **OBJECTIVES:** Enhance clarity, simplify information, improve color and data presentation without any bias, and engage the viewer.
- **DATA SOURCE:** By looking at the following data I have manually typed each and each category and sub-category in an Excel spreadsheet which allows me to run on Tableau.

	A	B	C	D
1	CATEGORY	SUB CATEGORIES	2018 (BILLIONS OF 2018 DOLLARS)	CHANGE FROM 2017
2	HEALTH	MEDICAID	93.3	1.6
3	HEALTH	CHIP	16.4	0.6
4	HEALTH	VACCINES FOR CHILDREN	4.4	-0.1
5	HEALTH	OTHER HEALTH	2.1	-0.1
6	NUTRITION	SNAP (formerly food stamps)	29.7	-1.4
7	NUTRITION	CHILD NUTRITION	22.7	-0.1
8	NUTRITION	SPECIAL SUPPLEMENTAL FOOD (WIC)	4.8	-0.3
9	INCOME SECURITY	SOCIAL SECURITY	21	-0.2
10	INCOME SECURITY	TEMPORARY ASSISTANCE FOR NEEDY FAMILIES	13.4	0.3
11	INCOME SECURITY	SUPPLEMENTAL SECURITY INCOME	9.7	-1.1
12	INCOME SECURITY	VETERANS BENEFITS	7.1	-0.2
13	INCOME SECURITY	CHILD SUPPORT ENFORCEMENT (net)	3.5	0
14	INCOME SECURITY	OTHER INCOME SECURITY	0	0
15	EDUCATION	EDUCATION FOR DISADVANTAGED (Title 1, Part A)	15.3	-1.3
16	EDUCATION	SPECIAL EDUCATION/IDEA	12.9	0
17	EDUCATION	SCHOOL IMPROVEMENT	4.2	-0.3
18	EDUCATION	IMPACT AID	1.5	-0.1
19	EDUCATION	INDIAN EDUCATION	1.3	0.1
20	EDUCATION	DEPENDENTS SCHOOLS ABROAD	1.2	0
21	EDUCATION	INNOVATION AND IMPROVEMENT	1.1	-0.1
22	EDUCATION	OTHER EDUCATION	2.9	-0.3
23	EARLY EDUCATION AND CARE	HEAD START (including early head start)	8.9	-0.2
24	EARLY EDUCATION AND CARE	CHILD CARE AND DEVELOPMENT FUND	5.9	0.1
25	EARLY EDUCATION AND CARE	OTHER EARLY EDUCATION AND CARE	0.3	0
26	SOCIAL SERVICES	FOSTER CARE	5.3	0.3
27	SOCIAL SERVICES	ADOPTION ASSISTANCE	2.9	0.4
28	SOCIAL SERVICES	UNACCOMPANIED ALIEN CHILDREN	1.2	-0.2
29	SOCIAL SERVICES	OTHER SOCIAL SERVICES	2.8	0
30	HOUSING	SECTION 8 LOW-INCOME HOUSING ASSISTANCE	7.7	0.1
31	HOUSING	LOW-RENT PUBLIC HOUSING	1	0
32	HOUSING	OTHER HOUSING	0.7	0
33	TRAINING	TRAINING	1.1	-0.2
34	REFUNDABLE PORTION OF TAX CREDITS	EARNED INCOME TAX CREDIT	52.1	-2.2
35	REFUNDABLE PORTION OF TAX CREDITS	CHILD TAX CREDIT	18.6	-1.2
36	REFUNDABLE PORTION OF TAX CREDITS	PREMIUM TAX CREDIT	1.1	0.5
37	REFUNDABLE PORTION OF TAX CREDITS	OTHER REFUNDABLE TAX CREDITS	0.9	0
38	TAX REDUCTIONS	CHILD TAX CREDIT (non refundable)	54.9	24.9
39	TAX REDUCTIONS	EXCLUSION FOR EMPLOYER-SPONSORED HEALTH INSURANCE	25.1	-0.3
40	TAX REDUCTIONS	DEPENDENT EXEMPTION	10.1	-31.1
41	TAX REDUCTIONS	EARNED INCOME TAX CREDIT (non refundable)	6.8	-0.1
42	TAX REDUCTIONS	DEPENDENT CARE CREDIT	3.6	-0.1
43	TAX REDUCTIONS	OTHER TAX REDUCTIONS	5.1	-0.3
44	TOTAL	TOTAL	484.6	-12.6

- **SOFTWARE SOLUTION:**
I created a good graph using Tableau after saving the mentioned data as a CSV file.


Federal Expenditures on Children in 2018
Category
Sub Categories
EARLY EDUCATION AND CARE
HEAD START (including early head start) 8.9 -0.20
CHILD CARE AND DEVELOPMENT FUND 5.9 0.10
OTHER EARLY EDUCATION AND CARE 0.3 0.00
EDUCATION
EDUCATION FOR DISADVANTAGED (Title 1, Part A) 15.3 -1.30
SPECIAL EDUCATION/IDEA 12.9 0.00
SCHOOL IMPROVEMENT 4.2 -0.30
OTHER EDUCATION 2.9 -0.30
IMPACT AID 1.5 -0.10
INDIAN EDUCATION 1.3 0.10
DEPENDENTS SCHOOLS ABROAD 1.2 0.00
INNOVATION AND IMPROVEMENT 1.1 -0.10
HEALTH
MEDICAID 93.3 1.60
CHIP 16.4 0.60
VACCINES FOR CHILDREN 4.4 -0.10
OTHER HEALTH 2.1 -0.10
HOUSING
SECTION 8 LOW-INCOME HOUSING ASSISTANCE 7.7 0.10
LOW-RENT PUBLIC HOUSING 1.0 0.00
OTHER HOUSING 0.7 0.00
INCOME SECURITY
SOCIAL SECURITY 21.0 -0.20
TEMPORARY ASSISTANCE FOR NEEDY FAMILIES 13.4 0.30
SUPPLEMENTAL SECURITY INCOME 9.7 -1.10
VETERANS BENEFITS 7.1 -0.20
CHILD SUPPORT ENFORCEMENT (net) 3.5 0.00
OTHER INCOME SECURITY 0.0 0.00
NUTRITION
SNAP (formerly food stamps) 29.7 -1.40
CHILD NUTRITION 22.7 -0.10
SPECIAL SUPPLEMENTAL FOOD (WIC) 4.8 -0.30
REFUNDABLE PORTION OF TAX CREDITS
EARNED INCOME TAX CREDIT 52.1 -2.20
CHILD TAX CREDIT 18.6 -1.20
PREMIUM TAX CREDIT 1.1 0.50
OTHER REFUNDABLE TAX CREDITS 0.9 0.00
SOCIAL SERVICES
FOSTER CARE 5.3 0.30
ADOPTION ASSISTANCE 2.9 0.40
OTHER SOCIAL SERVICES 2.8 0.00
UNACCOMPANIED ALIEN CHILDREN 1.2 -0.20
TAX REDUCTIONS
CHILD TAX CREDIT (non refundable) 54.9 24.90
EXCLUSION FOR EMPLOYER-SPONSORED HEALTH INSURAN.. 25.1 -0.30
DEPENDENT EXEMPTION 10.1 -31.10
EARNED INCOME TAX CREDIT (non refundable) 6.8 -0.10
OTHER TAX REDUCTIONS 5.1 -0.30
DEPENDENT CARE CREDIT 3.6 -0.10
TRAINING
TRAINING 1.1 -0.20
TOTAL
TOTAL 484.6 -12.60
0 50 100 150 200 250 300 350 400 450 500
2018 (Billions Of 2018 Dollars)
-100 -80 -60 -40 -20 0 20 40 60
Change From 2017
Category
EARLY EDUCATION A..
EDUCATION
HEALTH
HOUSING
INCOME SECURITY
NUTRITION
REFUNDABLE PORTI..
SOCIAL SERVICES
TAX REDUCTIONS
TRAINING
TOTAL
expenditures 2018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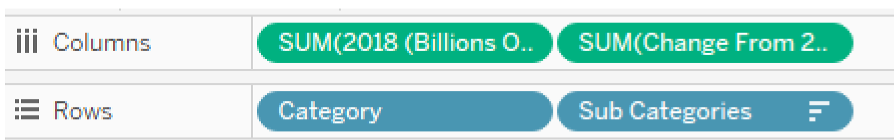
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Category
Sub Categories


Category
EARLY EDUCATION A..
EDUCATION
HEALTH
HOUSING
INCOME SECURITY
NUTRITION
REFUNDABLE PORTI..
SOCIAL SERVICES
TAX REDUCTIONS
TOTAL
TRAINING

- **COGNITIVE-TESTING QUESTIONS:**

  I. Based on the bar chart, can you identify which category had the highest expenditure in 2018?

  - The highest expenditure in 2018 based on the bar chart is the health category as it has the highest value compared to others.

  II. What category showed the most significant change from 2017 to 2018?

  - Tax reduction has shown the most significant changes from 2017 to 2018 as the values went in minus.

  III. Why is it important to choose distinct and meaningful colors for each category in a graph like this?

  - For Comparative Analysis, Easy Reference, Clarity, and Interpretability it is important to choose distinct and meaningful colors for each category in a graph.

  IV. How might this chart be useful for decision-making within an organization?

  - It has distinct colour and proper axis parameters, also the changes chart helps in analysis which helps to predict insights and helps in decision making.

  V. If you were tasked with further exploring this data, what additional visualizations or analyses might you consider gaining deeper insights?

  - According to me, the graph is perfect, I would do the same plotting.

  These are the responses by the reviewer, he has answered all the questions, agreed with my graph, and said the graph is perfect.

- **CHALLENGES:**

  The difficulties I encountered while creating a good graph are that, when we look at a bad graph, there is no comparison between the years 2017 and 2018. Since it was already mentioned in the data, I decided to include it. The values used to overlap when I tried to label each subcategory value. It worked when I decided to present both graphs separately and label them accordingly.

- **REDESIGNED GRAPHS IMPROVEMENTS:**

  After considering all its weaknesses, I have created a much simpler version of the original graph. Now, every category and subcategory is clear to the audience when they look at it. The graph now represents the data in a fair neutral manner and is no longer biased. This redesign ensures that viewers can quickly grasp the information without any confusion.

- **PATTERNS IN REDESIGNED GRAPH:**

Several important principles are at work when converting a poorly designed graph into a useful and appealing representation. First and foremost, simplicity and clarity are crucial. To reduce distractions, unnecessary components, clutter, and excessive decorations must be eliminated. For example, using bar charts for quantity comparison aligns the visualization with the nature of the data. Choosing the right chart type is crucial. They are labeling everything clearly, from axes to data points, and using color distinctions wisely. Without reduction, the scaling of the axes should accurately reflect the scale of the data. Organizing data logically with clear titles and captions guide the viewer. A well-designed graph also makes good use of consistency, accuracy, and accessibility.

4. **CONCLUSION AND POSSIBLE ALTERNATIVE DESIGN APPROACHES:**

In conclusion, when examining this independent dataset with distinct values for each category and sub-category, my choice of a bar graph aligns perfectly with the data's nature. It offers a straightforward and comprehensible way to present the information. I opted for this graph due to its suitability for the dataset, and I see no need for alternative approaches, as it effectively encapsulates the data in a user-friendly and visually appealing manner, aligning with my initial considerations.

5. **REFERENCES:**

- Budgeting for the Children: Where Does the Money Go? (howmuch.net)

The bad graph and the data required for the graph are extracted from the above website.

- OpenAI. (2023). ChatGPT. San Franciso: OpenAI.

Took some help to correct the grammatical mistakes.